Lec 09 环 Ring

$(R,+,\cdot)$

- $(R,+)$ 是阿贝尔群
  对加法：结合/交换律
  ⊕零元
- $(R,\cdot)$ semigroup
  对乘法：结合律

- Distributive: $a(b+c)=ab+ac$ $(b+c)a=ba+ca$

Rng 环 Ring
Ring 含幺环 Ring with 1

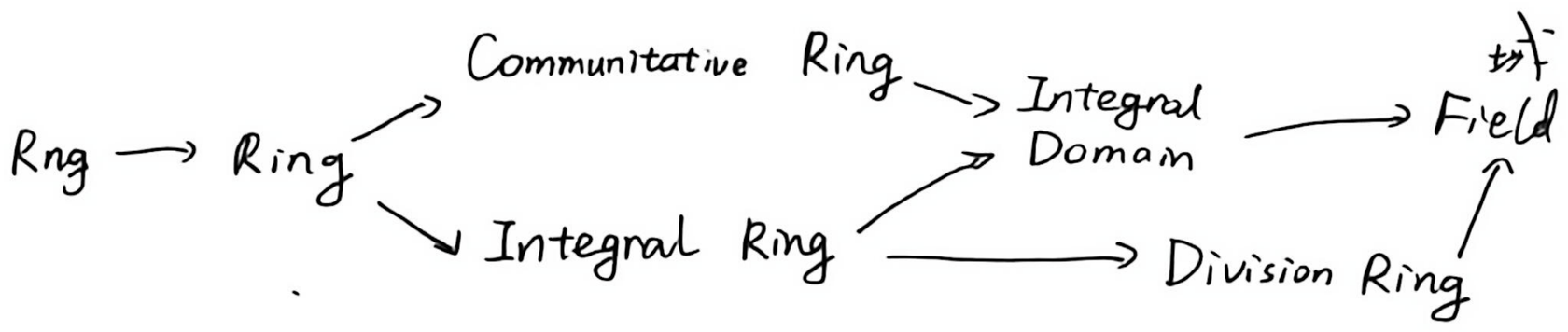
- Unital 1 有单位元 $1\neq 0$ $1\cdot a=a\cdot 1=a$

Commutative Ring 交换环：乘法是交换的

Devision Ring: $\forall a\neq 0, \exists a^{-1}.\ a\cdot a^{-1}=1$ → Field 域

zero factor: $a,b\in R\setminus\{0\}$ $ab=0$ a是左零因子 b是右零因子

Integral: no zero factor



Eg $\mathbb{Z}$ $\mathbb{Q}$ $\mathbb{R}$ $\mathbb{Z}_n=\mathbb{Z}/n\mathbb{Z}$

For Abel M 自同态 End M

$Matr(\mathbb{R})$ = $n\times n$ matrix $\mathbb{R}$

Product $\mathbb{R}_1\times\mathbb{R}_2\times\mathbb{R}_3$

Function Ring R^S $(f+g)(s)=f(s)+g(s)$ $(f\,g)(s)=f(s)\,g(s)$

Polynomial Ring $R[x]$ $f(x)=c_0+c_1x+c_2x^2+\cdots+c_dx^d$

Group Ring $R[G]$ semigroup $\left(\sum_{g\in G}a_g\cdot g\right)\left(\sum_{g\in G}b_g\cdot g\right)=\sum_g\left(\sum_{h_1,h_2=g}a_{h_1}b_{h_2}\right)g$

$\sum_{g\in G}a_g\cdot g\in R[G]$

Real Hamiltor Quaternums $\{\pm 1,\pm i,\pm j,\pm k\}$

$i^2=j^2=k^2=-1$, $ij=k$, $ji=-k$ $jk=i$ $ki=j$ $a+bi+cj+dk$
$kj=-i$ $ik=-j$ $a,b,c,d\in\mathbb{R}$

不是交换环，但是是 division ring (每个元可找到逆元)

Subring 子集 + 同样运算下构成环

Homomorphism 同态 $f: R\to R'$

$f(a+b)=f(a)+f(b)$
$f(ab)=f(a)\,f(b)$

Inf is a subring of R'
Kerf is a subring of R → an Ideal

若 $a\in R$, $b\in \ker f$ → $ab\in(\ker f)$ Subring
$f(ab)=f(a)\cdot\underline{f(b)}=0$ $ba\in\ker f$. Ideal 理想
0

(I,+) is a subgroup $\forall a\in I, b\in R$ $ab, ba\in I$

商环 $R/I = \{a+I\}$ $(a+I)+(b+I) \overset{def}{=} (a+b)+I$

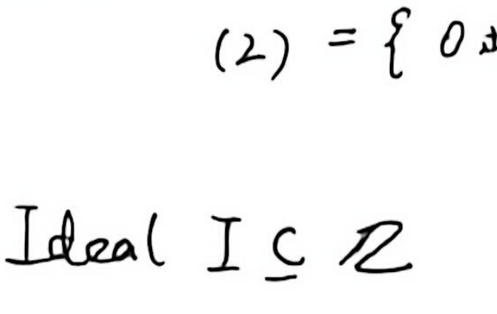
$(a+I)(b+I) \overset{def}{=} ab+I$

良定义 $a+I=a'+I$ $b+I=b'+I \Rightarrow ab+I=a'b'+I$

Pf. $a'=a+i$ $i\in I$

$b'=b+j$ $j\in I$

$a'b'=ab+\underbrace{aj+ib+ij}_{\text{和}\in I}$



$\tau(a)=a+I$

$\bar{f}(a+I)=f(a)$

E.g. I, J are Ideals $\longrightarrow$ $I+J$ is an Ideal

$I \cap J$ is an Ideal

For $S \subseteq R$ (S) = smallest ideal containing S = Ideal generated by S

$= \{\sum_{i=1}^{n} x_i a y_i\} = \{\sum x_i a y_i\} = \{\sum x_i a_i\}$ if commutative

$x, y \in R$, $a \in S$

Principal ideal 主理想

$(a) = (\{a\}) = \{\sum_{i=1}^{n} x_i a y_i\}$ if commutative $= \sum_{i=1}^{n} x_i a_i$

$\mathbb{Z}$ 上的理想

$(2) = \{0, \pm 2, \pm 4, \cdots\} \subset \mathbb{Z}$

Ideal $I \subseteq \mathbb{Z}$ $d = gcd(I)$ $d \in I$ $(d) \subseteq I$

$(a) \cap (b) = (lcm(a,b))$

$(a)+(b) = (gcd(a,b))$

(ab) $(a)(b) = (ab)$ $(a)(b)$ 元素乘积生成的 Ideal

$R \to R/I$ $a \to a+I$

any ideal I, J s.t. $I \subseteq J$ $R/J \cong (R/I)/(J/I)$

Pf. $f: R/I \to R/J$ 是满的. 且其核为 J/I. 故 $\cong$

满同态: $f: R \to R'$

$\{$any ideal of R containing $\ker f\} \longleftrightarrow \{$any ideal of $R'\}$

trivial Ideal $\{0\}$ R

① simple ring = no trivial Ideal

② commutative group + simple = field

考虑 $\forall a \in R/\{0\}$ $(a)=R$ $\{xa \mid x \in R\} \ni 1$ $\therefore a$ 可逆

$\therefore$ 交换 + 可逆 $\longrightarrow$ Field

③ Maximum ideal 极大理想 (非平凡) 没有非平凡理想

commutative ring + maximum ideal $\longrightarrow$ R/I is a field

④ prime ideal $\forall a, b$ $ab \in I \to a \in I$ or $b \in I$

commutative ring + prime ideal $\longrightarrow$ R/I integral

Pf. $(a+I)(b+I)=(ab+I)=0$

commutative ring R

I, J are (co)prime if $I+J=R$ $\quad (a)+(b)=\gcd(a,b)$

I, J are coprime $\Rightarrow I\cap J = IJ$

Pf. $IJ \subseteq I\cap J$ trivial

$1 \in I+J \quad 1 = i+j \in I+J$

$\forall a \in I\cap J \quad a = a(i+j) = ai+aj$

$\in IJ$

$R/IJ \cong R/I \times R/J$

Pf: $f: R \to R/I \times R/J \quad x \mapsto (x+I, x+J)$

$\ker f = I\cap J \quad 1 = i+j \in I+J$

$(a+I, b+J) \quad (bi+aj) \mapsto (a+I, b+J)$

$a(1-i)$

unit u if $\exists v$

$uv = vu = 1$

In commutative ring:

r associates s if $r = us$ for some unit

$r \sim s$

$R^* = \{\text{unit of } R\}$

Integral Domain

fraction field

$\frac{a \in R}{b \in R\setminus\{0\}}$

$R\times R\setminus\{0\}/\sim$

$=$

$\mathrm{Frac}(R)$

eg. $\mathbb{Z}$ $\quad \mathbb{Z}\setminus\{0\}$ $\quad \frac{a}{b}$

两个分数相等 $\frac{a}{b}\sim\frac{c}{d} \Leftrightarrow ad=bc$

$\mathbb{F}[x]$ $\quad c_0+c_1x+\cdots+c_dx^d$ $\quad \frac{c_0+c_1x+\cdots}{b_0+b_1x+\cdots}$

Characteristic $\underbrace{1+1+\cdots+1=0}_{\text{Characteristic}}$ iff $\mathrm{char}(R) \mid n$

$\mathbb{Z}/p\mathbb{Z}$ $\quad \mathrm{Frac}(\mathbb{Z}/p\mathbb{Z}[x])$